



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2025-26(Even Semester)

| VISION                                                                                       | MISSION                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To emerge as a learning hub and center of excellence in the domain of Information Technology | <p>[M1] To impart quality technical education through effective teaching learning process.</p> <p>[M2] To provide a platform to address societal issues as well as challenges faced by IT industries.</p> <p>[M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT.</p> <p>[M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.</p> |

### Class Test (CT)-1

Semester: VI

Programme: B. Tech Information Technology

Course: Ethical Hacking

Time: 1 Hr

Max Marks: 20 Marks

#### Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

| BIT33603 | Course Outcome                                                                                                                     |
|----------|------------------------------------------------------------------------------------------------------------------------------------|
| CO1      | To study the fundamental concepts of ethical hacking, including various phases and types of hackers                                |
| CO2      | To identify threats, vulnerabilities, and attacks in computer systems and networks                                                 |
| CO3      | To explore tools and techniques used in ethical hacking such as foot printing, scanning, social engineering, and password cracking |
| CO4      | To evaluate system and network security measures using penetration testing and vulnerability assessment tools                      |
| CO5      | Demonstrate awareness of the legal, ethical, and professional issues related to cybersecurity practices                            |

| Group A |                               |     |     |       |
|---------|-------------------------------|-----|-----|-------|
| Que. 1  | Answer the following Question | CO  | BTL | Marks |
| a.      | Define social engineering.    | CO1 | 3   | 2     |
| b.      | What is insider threat?       | CO2 | 2   | 2     |

| Group B |                                               |     |     |       |
|---------|-----------------------------------------------|-----|-----|-------|
| Que. 2  | Answer the following Question (Solve any Two) | CO  | BTL | Marks |
| a.      | Explain scanning techniques                   | CO1 | 4   | 4     |
| b.      | Describe footprinting methods                 | CO1 | 5   | 4     |

|        |                                                   |     |     |       |
|--------|---------------------------------------------------|-----|-----|-------|
| c.     | Explain types of ethical hacking.                 | C01 | 4   | 4     |
| Que 3. | Answer the following Question (Solve any Two)     | CO  | BTL | Marks |
| a.     | Explain OS vulnerabilities.                       | CO2 | 4   | 4     |
| b.     | Describe network-based attacks.                   | CO2 | 5   | 4     |
| c.     | Explain security challenges in wireless networks. | CO2 | 4   | 4     |

**Bloom's Taxonomy Level (BTL)-** Remember , Understand , Apply, Analyze , Evaluate , Create